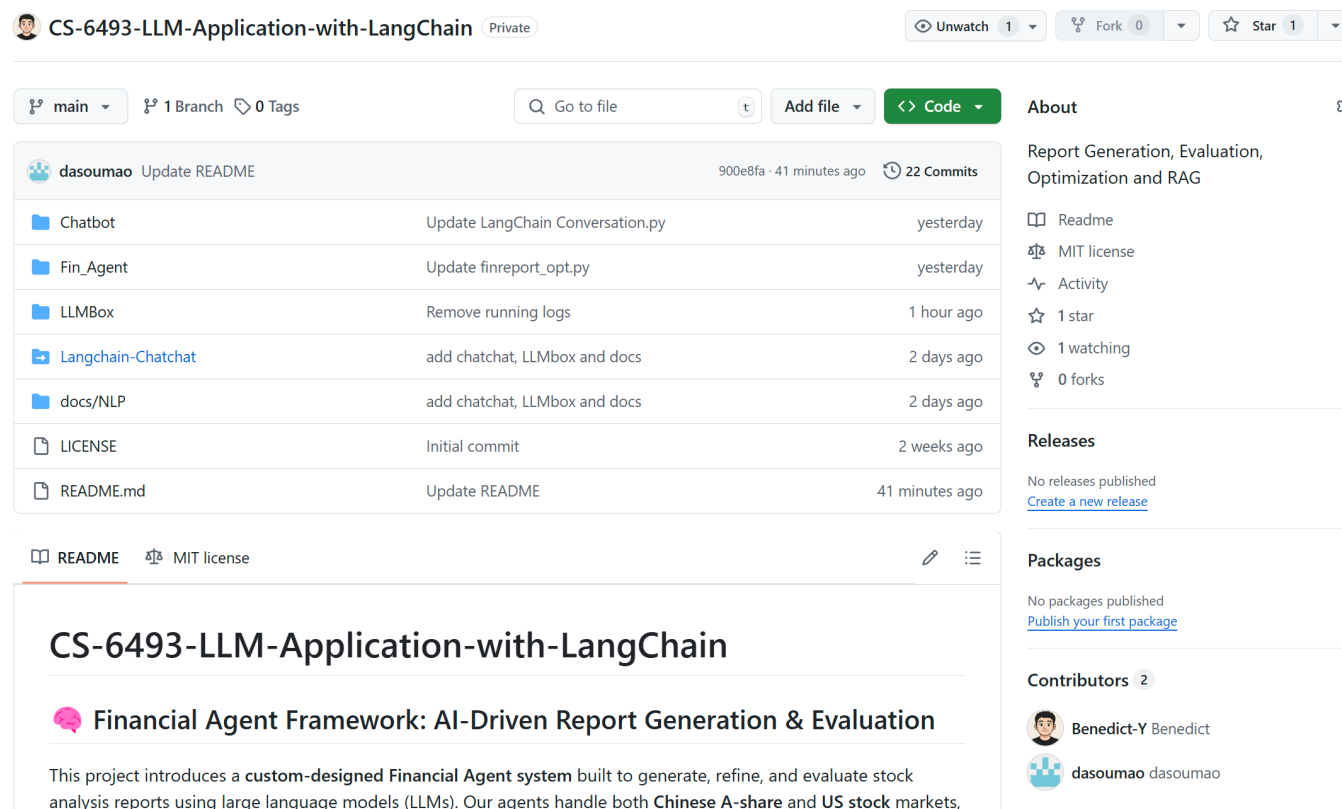


Financial Report Generation and Optimization System Based on LangChain

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Postgraduate, CityU-DG
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CS-6493-LLM-Application-with-LangChain Private

Unwatch 1 Fork 0 Star 1

main 1 Branch 0 Tags

Go to file Add file Code

File/Folder	Commit Message	Time Ago
dasoumao	Update README	900e8fa · 41 minutes ago 22 Commits
Chatbot	Update LangChain Conversation.py	yesterday
Fin_Agent	Update finreport_opt.py	yesterday
LLMBox	Remove running logs	1 hour ago
Langchain-Chatchat	add chatchat, LLMbox and docs	2 days ago
docs/NLP	add chatchat, LLMbox and docs	2 days ago
LICENSE	Initial commit	2 weeks ago
README.md	Update README	41 minutes ago

README MIT license

CS-6493-LLM-Application-with-LangChain

 **Financial Agent Framework: AI-Driven Report Generation & Evaluation**

This project introduces a custom-designed **Financial Agent** system built to generate, refine, and evaluate stock analysis reports using large language models (LLMs). Our agents handle both **Chinese A-share** and **US stock** markets,

<https://github.com/Benedict-Y/CS-6493-LLM-Application-with-LangChain>

1. Introduction and Related Work
2. Financial Report System Design
3. System Agents
4. Agent Results
5. Financial Report System
6. Conclusion and Future Work

Introduction and Related Work

Motivation

Market and Application Needs

- Investors require more objective and comprehensive financial reports to support decision-making.
- Financial institutions need to unify handling of **structured market data and unstructured news**.

Challenges of Applying LLMs in Finance

- **Domain Adaptation:** General models lack deep understanding of financial terms, regulations, and market mechanics.
- **Hallucination:** LLMs may produce fluent but factually incorrect outputs (“**hallucinations**”), which is risky for investment decisions.
- **System Integration Complexity:** Report generation involves complex stages—data acquisition, quantitative analysis, drafting, evaluation, and iterative optimization—that require smooth coordination among multiple modular agents.

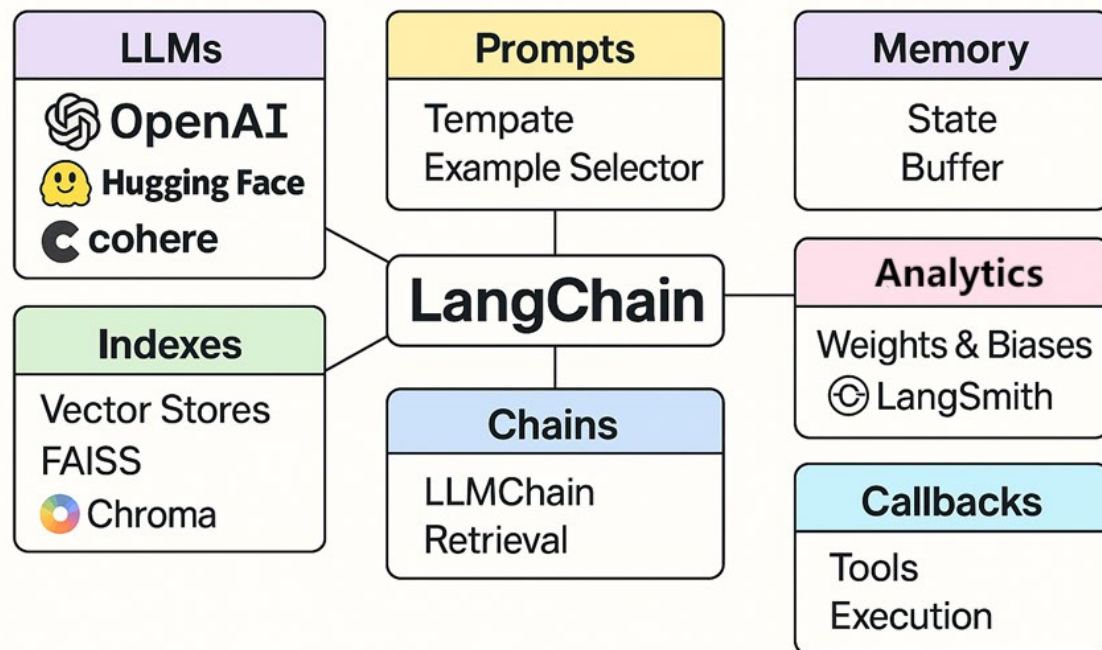


Introduction and Related Work



LangChain

LangChain Ecosystem



Core Modules of LangChain

LangChain provides the following 6 standardized and extensible core modules, with interfaces for integration:

1. Model I/O: Interfaces for interacting with language models.

2. Data Connection: Interfaces for interacting with data from specific applications.

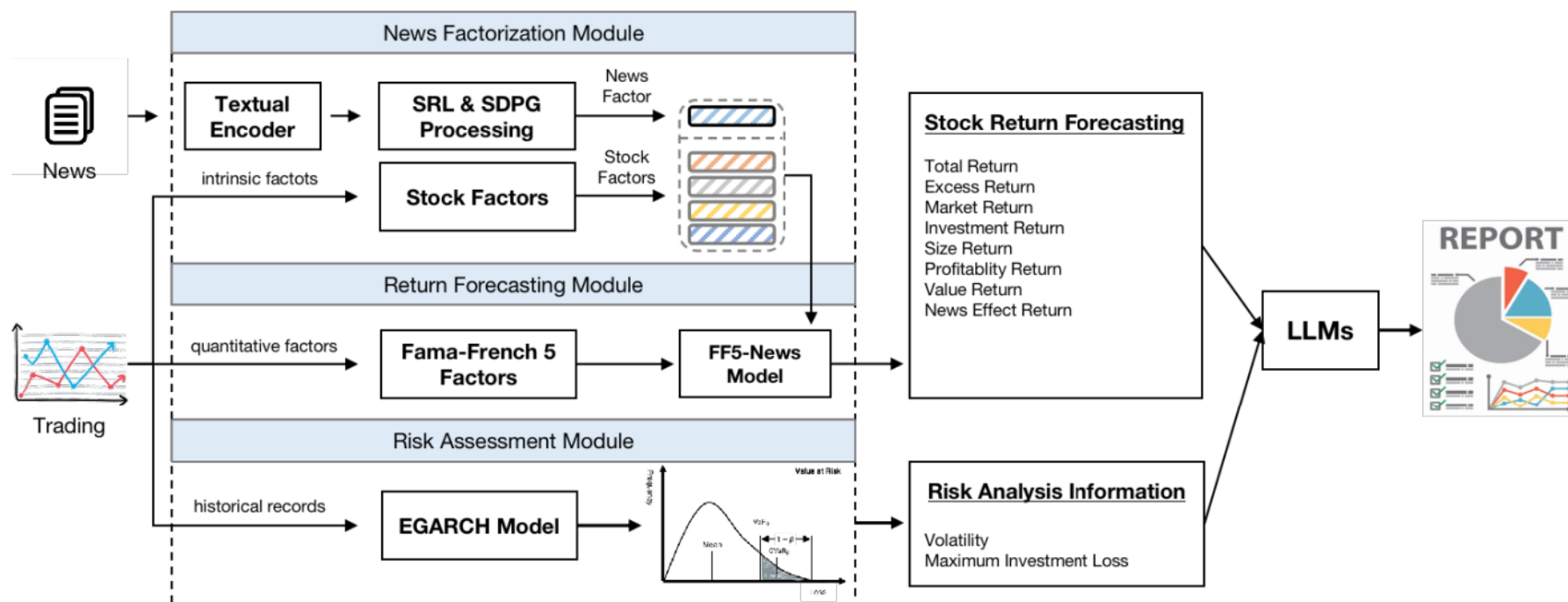
3. Chains: Used for orchestrating complex application workflows.

4. Agents: Use language models as reasoning engines to determine action sequences.

5. Memory: Maintains persistent application state across multiple runs of a chain.

6. Callbacks: Middleware steps to record and stream information within any chain components.

Introduction and Related Work



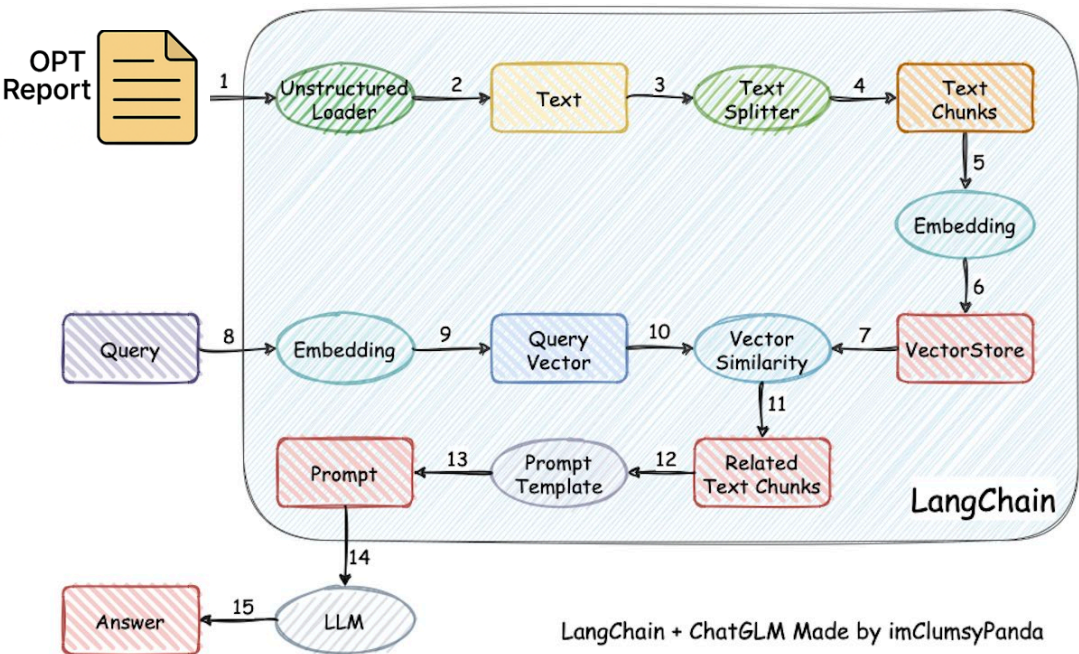
The **missing work** includes:

- No supplementary additions of **user-concerned content** to the generated reports
- No **hallucination detection and correction** for LLM-generated content
- No establishment of a database for **storing and querying** the series of generated reports

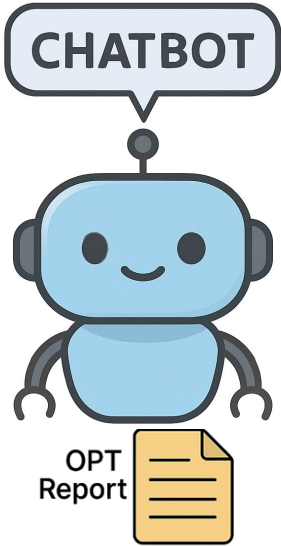
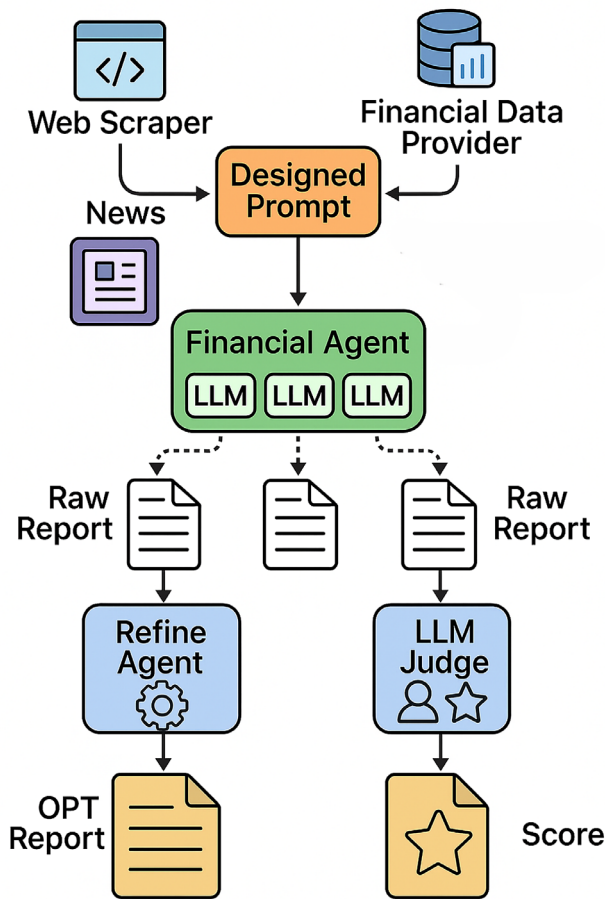
[WWW'2024] "FinReport: Explainable Stock Earnings Forecasting via News Factor Analyzing Model"

Financial Report System Design


 LangChain-Chatcat



Retrieval-Augmented Generation

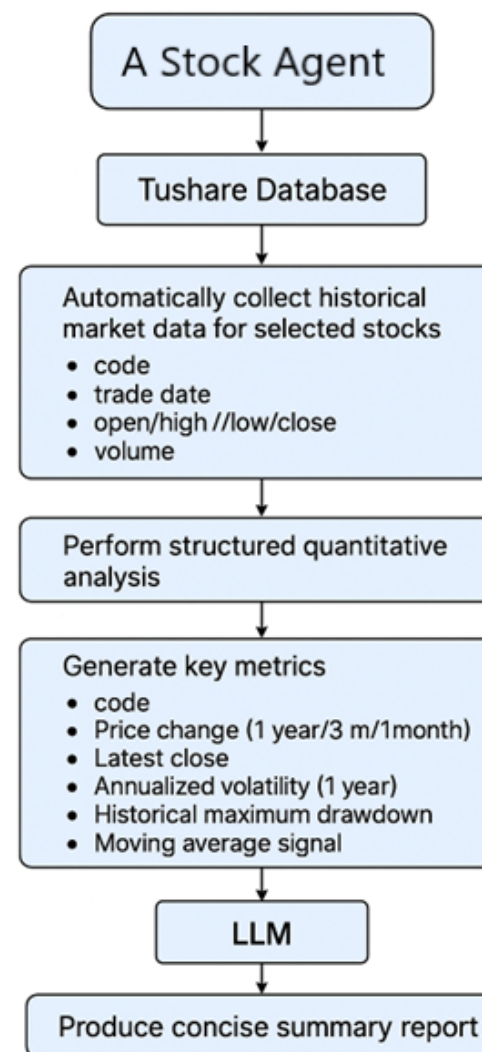
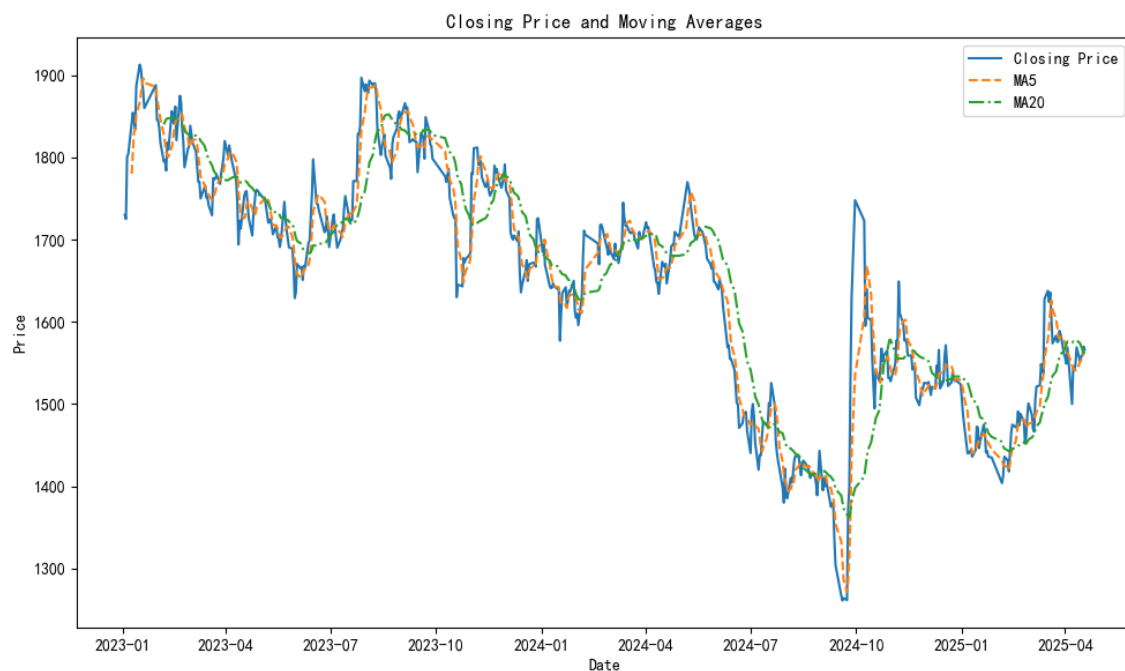


Conversation Buffer Memory

System Agent --- Financial Agents



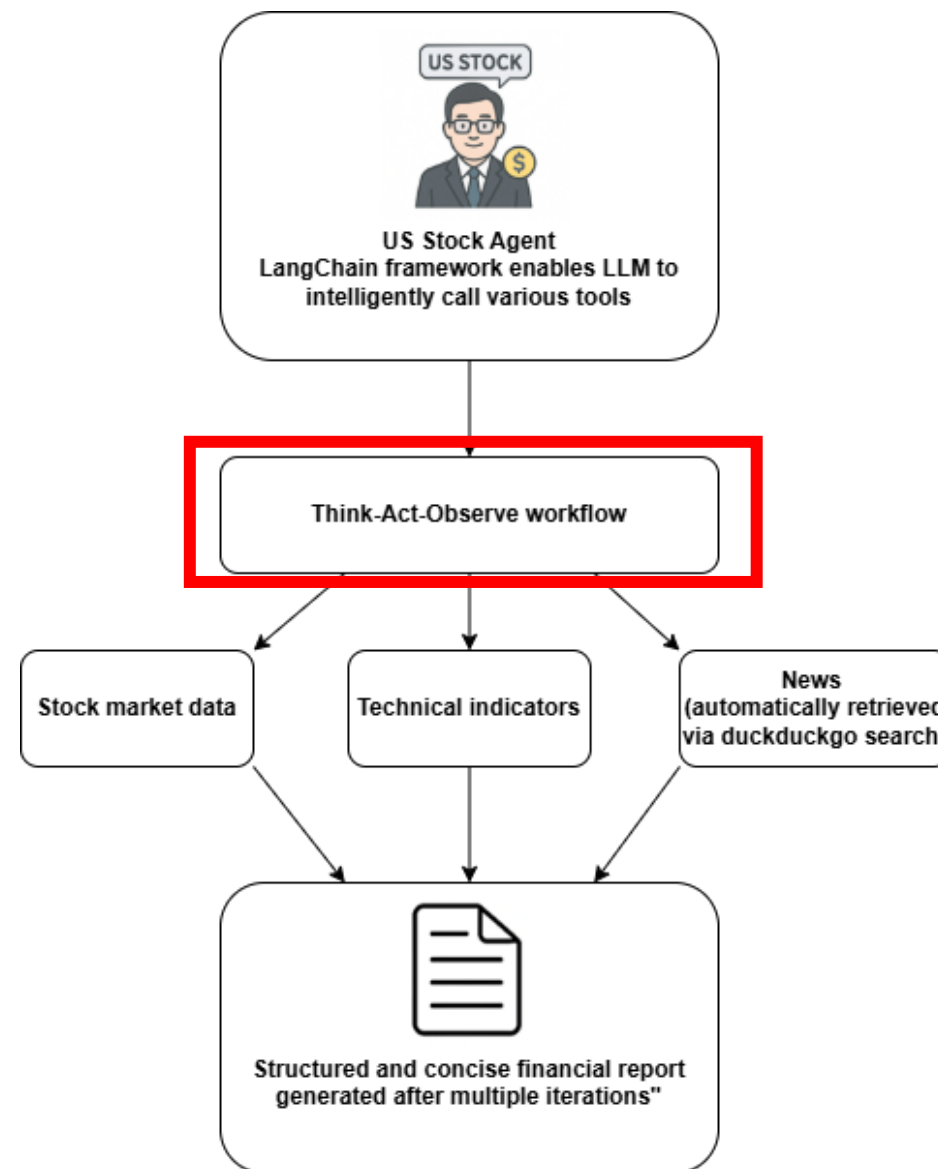
TSHARE



System Agent --- Financial Agents



LangChain



System Agent --- Refine Agent

贵州茅台(600519.SH)量化分析报告

报告日期: 2023年XX月XX日 🕒 Hallucination in Time

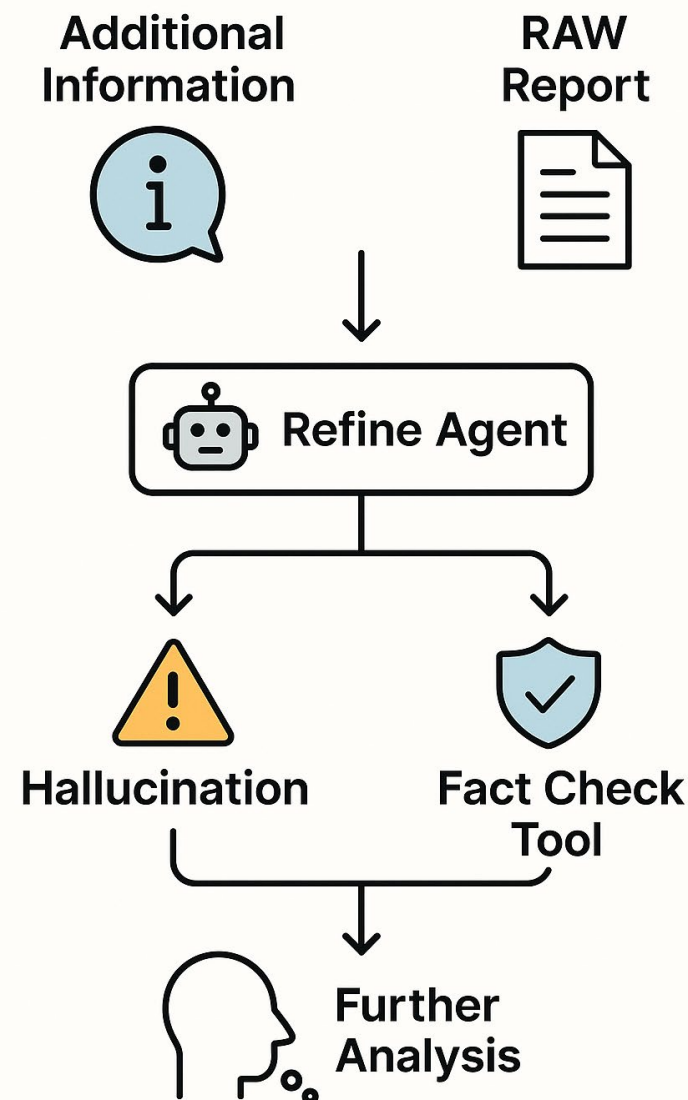
1. 行情回顾

📊 Hallucination in Data

- **长期表现:** 近一年累计收益 **-5.82%**, 显著弱于市场基准 (如沪深300同期涨幅X.X%)。
- **短期趋势:**
 - 近三月反弹**7.92%**, 显示阶段性修复;
 - 近一月回调 **-3.33%**, 或反映市场情绪波动。
- **绝对价格:** 最新收盘价**1570.00元**, 处于近一年价格区间[1400, 1900]的中低位。

(注: 均线系统未现明显交叉, 价格仍处于震荡整理阶段)

Solution



System Agent --- LLM Judge Agent

Challenges of Evaluating Results

- No Standard Answer
 - Financial Market Changing Rapidly
 - There are a thousand Hamlets in a thousand people's eyes.
- Ineffective Metrics
 - Similarity-based Metrics (ROUGE, BLEU, etc)
Do NOT Understand Report
- Expensive Human Evaluation
 - Financial Experts

CHALLENGES OF EVALUATING RESULTS



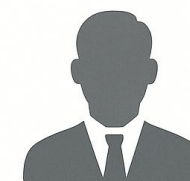
No Standard Answer



Ineffective Metrics



Ineffective Metrics



Expensive Human Evaluation



Zheng, Lianmin, et al. "Judging llm-as-a-judge with mt-bench and chatbot arena." Advances in Neural Information Processing Systems 36 (2023): 46595-46623.

System Agent --- LLM Judge Agent

Input Structure

- **Modularity (Structured)**

- Mandatory Reasoning (Chain-of-Thought)

- Clear Quantitative & Qualitative Evaluation

```
report_modules = [  
    "Market review",  
    "Risk characteristics",  
    "Technical indicators/signals",  
    "Recent News Analysis",  
    "Additional summary recommendations"  
]
```

https://huggingface.co/learn/cookbook/llm_judge

https://huggingface.co/learn/cookbook/structured_generation

System Agent --- LLM Judge Agent

Output Structure

- **Modularity (Structured)**

- Mandatory Reasoning (Chain-of-Thought)

- Clear Quantitative & Qualitative Evaluation

1. Overall Comparative Assessment:

1. Overall Reasoning
2. Overall Winner

2. Detailed Modular Comparison (Perform for EACH module listed above):

3. Strengths & Weaknesses Summary:

1. Report 1: Strengths - [...], Weaknesses - [...]
2. Report 2: Strengths - [...], Weaknesses - [...]

4. Detailed Scoring:

https://huggingface.co/learn/cookbook/llm_judge

https://huggingface.co/learn/cookbook/structured_generation

System Agent --- LLM Judge Agent

- Modularity (Structured)
- **Mandatory Reasoning (Chain-of-Thought)**
- Clear Quantitative & Qualitative Evaluation

Comparative Rationale (CoT): Provide a detailed side-by-side analysis comparing how each report handled this specific module based on the reference material and the evaluation aspects. Discuss specific examples.

Module Winner: State which report performed better for this specific module relative to the others and the reference.

Justification: Briefly explain the primary reason for the module winner decision, linking back to the rationale.

https://huggingface.co/learn/cookbook/llm_judge

https://huggingface.co/learn/cookbook/structured_generation

System Agent --- LLM Judge Agent

- Modularity (Structured)



Leave more time for thought by adding an **Evaluation field** before the final answer.



Use a **small integer scale** like 1-4 or 1-5 instead of a large float scale as we had previously.



Provide an **indicative scale** for guidance.

- Mandatory Reasoning (Chain-of-Thought)

• **Score Justification Logic:** Scores must reflect the *relative comparison* discussed in the modular analysis. A higher score indicates better performance *compared to the other report(s)* on that module

• **Scores:**

- Report 1:

- Market review: [Score 1-5] - Justification: [...]

- Risk character: [Score 1-5] - Justification: [...]

- **Overall Comparative Score:** [Score 1-5] - Justification: [Reflects overall standing relative to others based on reference usage]

- **Clear Quantitative & Qualitative Evaluation**

https://huggingface.co/learn/cookbook/llm_judge

https://huggingface.co/learn/cookbook/structured_generation

USStock

Financial Report for Apple Inc. (AAPL)

Date: Current

1. Company Overview and Stock Performance

- Current Stock Price: \$196.98
- Market Capitalization: \$2.96 trillion
- P/E Ratio: 31.27x
- Position: Maintains status as one of the world's most valuable public companies

2. Financial Analysis

Current Market Metrics:

- Premium valuation indicated by P/E ratio of 31.27x
- Market cap indicates strong investor confidence
- Note: Detailed financial statements were not available in our current data set

3. Recent News Analysis

Key Developments:

- Stock has recently traded below the \$3 trillion market cap level
- Company facing ongoing challenges related to tariffs and trade relations
- Continued focus on market expansion and product innovation
- Chinese market remains a crucial factor in company performance

4. Investment Outlook

Positive Factors:

- Strong market position in premium consumer electronics
- Robust ecosystem of products and services

AStock

Quantitative Analysis Report: 600519.SH

1. Market Performance Review

Price Performance

- Current Price: ¥1,565.94
- Period Returns:
 - 1-year return: -6.27%

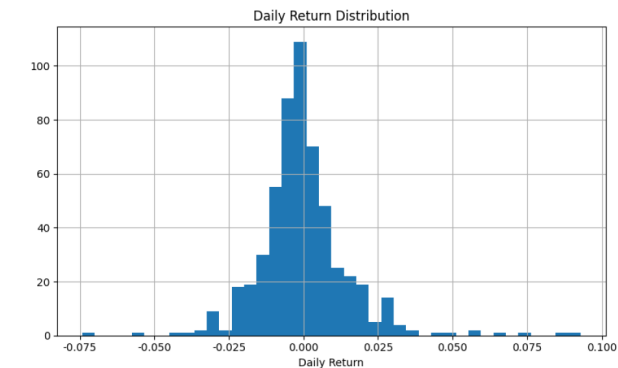
Key Observations

- The stock has shown mixed performance across different timeframes

2. Risk Characteristics

Drawdown Analysis

- Maximum Historical Drawdown: -34.08%
 - Represents substantial downside risk



USStock

```
---  
## 5. Risk Factors  
- **Supply Chain Disruptions**: Reliance on China/Taiwan for components.  
- **Competition**: Rising competition in wearables (e.g., Samsung) and services (Spotify, Netflix).  
- **Currency Fluctuations**: Strong USD could hurt international revenue.  
---  
  
**Data Sources**: Yahoo Finance, Xueqiu, Futu, East Money.  
**Disclaimer**: Always verify financial statements from official SEC filings before making investment decisions.  
Invalid Format: Missing 'Action:' after 'Thought:It seems there was a formatting error in my previous response. Let me correct that and proceed to generate the financial report properly.  
---  
  
# Financial Report for Apple Inc. (AAPL)  
## Date: October 2023  
---  
  
## 1. Company Overview and Stock Performance  
- **Current Stock Price**: $196.98  
- **Market Capitalization**: $2.96 trillion  
- **P/E Ratio**: 31.27  
  
Apple remains the world's most valuable publicly traded company, with its stock trading near all-time highs. The elevated P/E ratio suggests investors expect continued growth, though it may indicate overvaluation if earnings growth slows.  
---  
  
## 2. Financial Analysis  
*Note: Detailed financial statements (revenue, net income, etc.) were unavailable via the tools. For accurate metrics, refer to Apple's latest SEC filings (10-Q or 10-K).*  
---  
  
## 3. Recent News Analysis  
Key updates from financial platforms:  
1. **iPhone 16e Launch**: Apple captured 19% of global smartphone market share in Q1 2023, driven by strong demand for its new iPhone 16e and growth in emerging markets (Counterpoint Research via Futu).
```

AStock

Quantitative Analysis Report: 600519.SH

1. Market Performance Review

Price Performance

- Current Price: ¥1,565.94
- Period Returns:
 - 1-year return: -6.27%

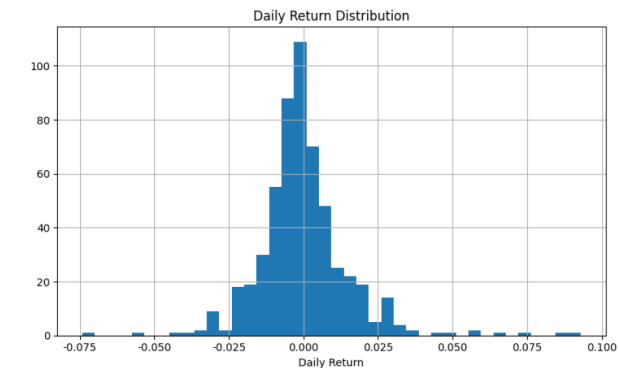
Key Observations

- The stock has shown mixed performance across different timeframes

2. Risk Characteristics

Drawdown Analysis

- Maximum Historical Drawdown: -34.08%
 - Represents substantial downside risk



Results --- RAW Report V.S. OPT Report

RAW REPORT

Quantitative Financial Analysis Report for Stock Code 600519.SH

1. Market Review

The stock code 600519.SH has exhibited mixed performance over the past year. Below are the key quantitative metrics:

- 1-Year Price Change: -6.27%
- More Detailed Metrics

The negative 1-year price change indicates a decline in value over the longer term, while the positive 3-month change suggests a recent recovery phase. However, the 1-month decline indicates potential volatility or market correction.

[Price Trend and Moving Averages Image is Not Shown Here]

2. Risk Characteristics

Volatility

- 1-Year Annualized Daily Return Volatility: 28.38%

This level of volatility suggests that the stock experiences significant price fluctuations, which may be indicative of higher risk for investors.

Maximum Drawdown

- Historical Maximum Drawdown: -34.08%

The maximum drawdown reflects the largest observed loss from a peak to a trough, indicating that the stock has experienced substantial declines in the past. This level of drawdown is a critical risk factor for potential investors.

3. Technical Indicators/Signals

The absence of significant moving average crossovers suggests a lack of clear bullish or bearish momentum signals in the short to medium term. This may indicate a period of consolidation or indecision in the market.

[Daily Return Distribution Image is Not Shown Here]

4. Additional Summary Recommendations

Based on the analysis of the stock code 600519.SH, the following recommendations are made:

- **Caution Advised:** Given the negative 1-year price change and high volatility, investors should exercise caution when considering new positions in this stock.
- **Monitor for Recovery:** The recent positive 3-month price change may indicate a potential recovery. Investors should monitor this trend closely for signs of sustained upward momentum.
- **Risk Management:** Due to the historical maximum drawdown of -34.08%, it is advisable for investors to implement robust risk management strategies, including stop-loss orders or position sizing, to mitigate potential losses.
- **Technical Analysis:** Continue to observe moving averages for any future crossover signals that may provide clearer entry or exit points.

In conclusion, while the stock has shown some short-term recovery, the overall risk profile suggests that investors should remain vigilant and consider their risk tolerance before making investment decisions.

OPT REPORT

Quantitative Financial Analysis Report for Stock Code 600519.SH

1. Market Review

<Same AS Left>

2. Risk Characteristics

<Same AS Left>

3. Technical Indicators/Signals

<Same AS Left>

4. Additional Summary Recommendations

Based on the analysis of the stock code 600519.SH, the following recommendations are made:

- **Caution Advised:** Given the negative 1-year price change and high volatility, investors should exercise caution when considering new positions in this stock.
- **Monitor for Recovery:** The recent positive 3-month price change may indicate a potential recovery. Investors should monitor this trend closely for signs of sustained upward momentum.
- **Risk Management:** Due to the historical maximum drawdown of -34.08%, it is advisable for investors to implement robust risk management strategies, including stop-loss orders or position sizing, to mitigate potential losses.
- **Technical Analysis:** Continue to observe moving averages for any future crossover signals that may provide clearer entry or exit points.

5. Recent News Impact Analysis

美国对华增加125%关税

The recent news of the United States imposing a 125% tariff increase on Chinese goods could have several potential impacts on the stock code 600519.SH:

Potential Negative Impacts:

1. **Increased Cost of Exports:** If 600519.SH relies on exporting its products to the U.S., the increased tariffs could lead to higher costs and reduced competitiveness in the American market.

Some More Negative Impacts

Potential Positive Impacts:

1. **Domestic Market Focus:** The company might shift its focus to the domestic market, which could benefit from reduced competition from U.S. imports, potentially leading to increased market share.

Some More Positive Impacts

Overall, the impact of the increased tariffs will depend on the company's ability to adapt its strategies to mitigate potential costs and capitalize on new opportunities. Investors should keep an eye on the company's announcements regarding their response to these trade developments.

Results --- LLM Judge of FLLM and GLLM

1. Overall Comparative Assessment:

- **Overall Reasoning (CoT):** Both reports provide a comprehensive analysis of stock code 600519.SH, covering market performance, risk characteristics, technical indicators, and recommendations. Report 1 is structured with clear sections and provides a concise summary and actionable recommendations. Report 2 offers a detailed explanation of each aspect, with a focus on risk management and long-term outlook. Report 1 is more succinct and focused on practical trading strategies, while Report 2 provides a broader context and detailed risk assessment.
- **Overall Winner:** Report 2 is judged superior overall based on its internal quality and analytical presentation. It provides a more thorough risk assessment and long-term outlook, which adds depth to the analysis.

2. Detailed Modular Comparison:

- **Module Name:** Market Performance Review
 - **Comparative Rationale (CoT):** Report 1 presents the market performance with clear data points and observations, highlighting mixed performance and potential short-term recovery. Report 2 provides similar data but integrates volatility and drawdown information into the market review, offering a more comprehensive view of the stock's performance and risk profile.
 - **Module Winner:** Report 2
 - **Justification:** Report 2 integrates risk characteristics into the market review, providing a more holistic view of the stock's performance and potential risks.
- **Module Name:** Risk Characteristics
 - **Comparative Rationale (CoT):** Report 1 discusses volatility and drawdown with a focus on their implications for risk. Report 2 elaborates on these metrics, explaining their significance and combining them into a comprehensive risk assessment. Report 2's detailed explanation of the risk profile is more informative.
 - **Module Winner:** Report 2
 - **Justification:** Report 2 provides a more detailed and insightful analysis of risk characteristics, enhancing the reader's understanding of the stock's risk profile.
- **Module Name:** Technical Indicators/Signals
 - **Comparative Rationale (CoT):** Both reports note the absence of significant moving average crossovers. Report 1 briefly mentions consolidation, while Report 2 explains the implications of the absence of crossovers and discusses daily return distribution, offering additional insights into the stock's technical profile.
 - **Module Winner:** Report 2
 - **Justification:** Report 2 provides a more comprehensive analysis of technical indicators, including daily return distribution, which adds depth to the technical analysis.
- **Module Name:** Summary and Recommendations

- **Comparative Rationale (CoT):** Report 1 offers specific trading recommendations, including position sizing and risk management strategies. Report 2 provides broader recommendations, including risk management and timing considerations, and discusses long-term outlook. Report 1 is more focused on actionable trading strategies.
- **Module Winner:** Report 1
- **Justification:** Report 1 provides clear and actionable recommendations, making it more practical for traders seeking specific guidance.

3. Strengths & Weaknesses Summary (Internal):

- Report 1: Strengths - Clear structure, actionable recommendations; Weaknesses - Less detailed risk assessment.
- Report 2: Strengths - Comprehensive risk assessment, detailed analysis; Weaknesses - Less focus on specific trading strategies.

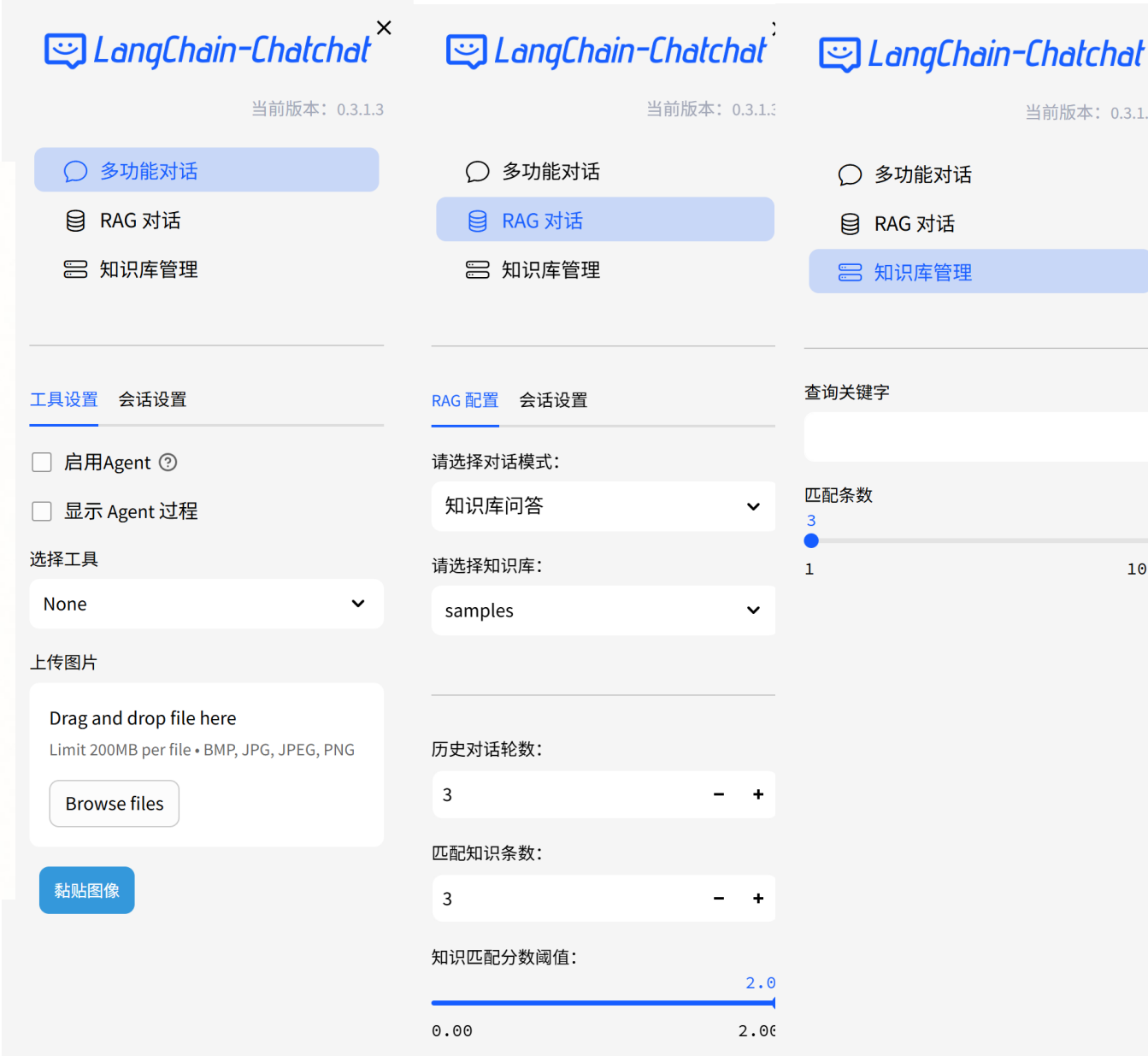
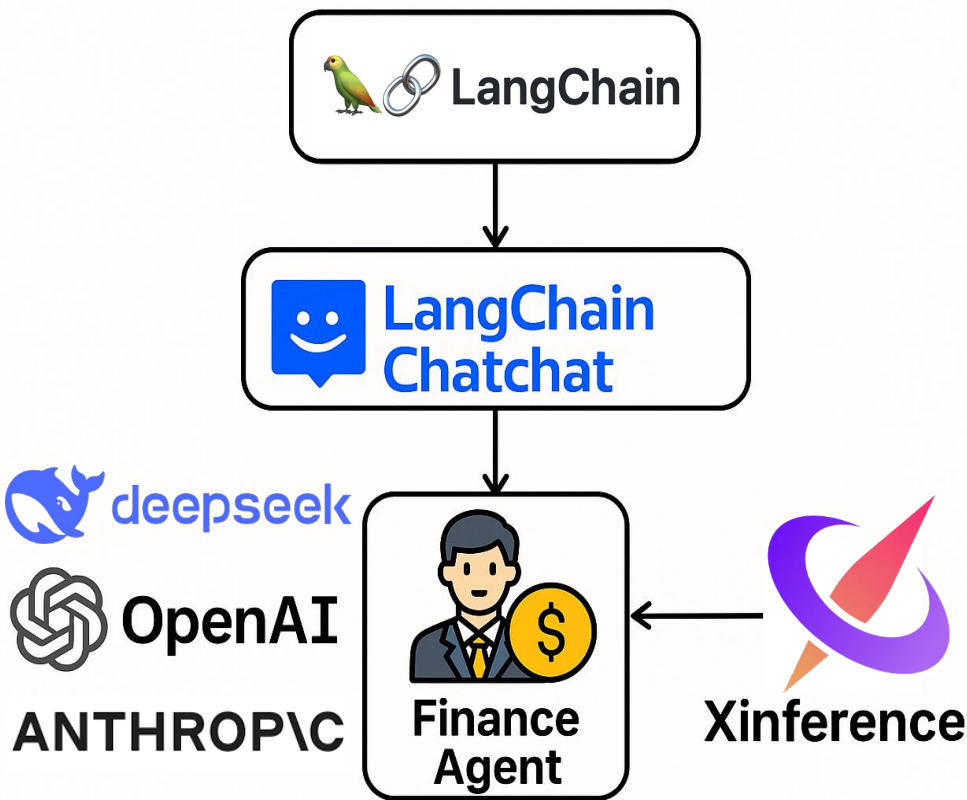
4. Detailed Scoring (Reflecting Internal Quality):

- **Scores:**
 - Report 1:
 - Market review: 3 - Provides basic performance data but lacks integration with risk characteristics.
 - Risk characteristics: 3 - Adequate discussion but less detailed than Report 2.
 - Technical indicators/signals: 3 - Brief analysis without additional insights.
 - Summary and Recommendations: 4 - Clear and actionable recommendations.
 - **Overall Comparative Score: 3** - Provides a practical analysis but lacks depth in risk assessment.
 - Report 2:
 - Market review: 4 - Integrates risk characteristics into market performance analysis.
 - Risk characteristics: 4 - Detailed and insightful risk assessment.
 - Technical indicators/signals: 4 - Comprehensive analysis including daily return distribution.
 - Summary and Recommendations: 3 - Broader recommendations but less specific trading guidance.
 - **Overall Comparative Score: 4** - Offers a thorough and insightful analysis with a focus on risk management.

Score Justification Logic: Scores reflect the relative comparison discussed in the modular analysis.

Report 2 consistently provides more detailed and insightful analysis, particularly in risk characteristics and technical indicators, while Report 1 excels in providing clear and actionable recommendations.

Financial Report System



Financial Report System --- Local Model Deploy



Fino1-8B

MEM: 92%

UTL: 85%

```
metrics.py:455] Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 52.2 tokens/s, Running: 1 reqs, Swapped: 0 req
GPU KV cache usage: 4.0%, CPU KV cache usage: 0.0%.
metrics.py:455] Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 52.3 tokens/s, Running: 1 reqs, Swapped: 0 req
GPU KV cache usage: 4.5%, CPU KV cache usage: 0.0%.
async_llm_engine.py:179] Finished request 81332522-1d8b-11f0-b292-5a3e0e0d5e9d.
async_llm_engine.py:211] Added request 88a87870-1d8b-11f0-b292-5a3e0e0d5e9d.
metrics.py:455] Avg prompt throughput: 435.1 tokens/s, Avg generation throughput: 48.5 tokens/s, Running: 1 reqs, Swapped: 0 r
s, GPU KV cache usage: 5.1%, CPU KV cache usage: 0.0%.
```



Running Models

LANGUAGE MODELS					EMBEDDING MODELS	RERANK MODELS	IMAGE MODELS	AUDIO MODELS
ID	Name	Address	GPU Indexes	Size				
llama-3.1-instruct	llama-3.1-i...	0.0.0.0:38607	0	8				



Fin-R1

MEM: 91%

UTL: 0%

```
async_llm_engine.py:211] Added request 0d8df0c4-1d8c-11f0-9dcf-5a3e0e0d5e9d.
metrics.py:455] Avg prompt throughput: 8.6 tokens/s, Avg generation throughput: 0.3 tokens/s, Running: 1 reqs, Swapped: 0 reqs
GPU KV cache usage: 0.6%, CPU KV cache usage: 0.0%.
metrics.py:455] Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 49.6 tokens/s, Running: 1 reqs, Swapped: 0 req
GPU KV cache usage: 1.2%, CPU KV cache usage: 0.0%.
metrics.py:455] Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 50.1 tokens/s, Running: 1 reqs, Swapped: 0 req
GPU KV cache usage: 1.6%, CPU KV cache usage: 0.0%.
metrics.py:455] Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 50.7 tokens/s, Running: 1 reqs, Swapped: 0 req
GPU KV cache usage: 2.2%, CPU KV cache usage: 0.0%.
metrics.py:455] Avg prompt throughput: 0.0 tokens/s, Avg generation throughput: 49.8 tokens/s, Running: 1 reqs, Swapped: 0 req
GPU KV cache usage: 2.7%, CPU KV cache usage: 0.0%.
```

Financial Report System --- Local Model Deploy

 Xinference Chat Bot : llama-3.1-instruct 

Model ID: llama-3.1-instruct
Model Size: 8 Billion Parameters
Model Format: pytorch
Model Quantization: none

Chatbot

hi

Hello! What's going on? Feel free to share anything you'd like to talk about or ask me if you need help with something. I'm here to chat!

Also, I noticed we had a previous conversation involving some text about dates and weather. If you'd like to continue where we left off or start fresh, just let me know!

Is there anything specific on your mind, or do you just want to see where the conversation takes us?

Thinking

Let's start with a simple hello until I understand where this conversation is headed. You seemed to already have something on your mind from our previous interaction. Let's see if we can dive back into that.

Oh, I got it! The previous text was about a date you mentioned, which fell on December 26, 2024. Now, what was next about that day? Aha, right, you asked about weather-related topics.

So, December 26th, 2024. Interesting. That's around the time when winter is supposed to be in full swing in the Northern Hemisphere, but I'm curious what specific weather phenomena you're thinking about.

Hmm, I'm getting a sense that you might be pondering the specifics of snow or maybe winter storms occurring around this date. Let me double-check—yes, December 2023 was indeed a year when snowy conditions were noted in many parts of the world.

 Retry

 Undo

 Clear

Type a message...

Submit

Additional Inputs

通过 API 使用  · 使用 Gradio 构建 



Financial Report System --- Chatbot Report Gen

Deploy

 **LangChain-Chatchat** ✕

当前版本: 0.3.1.3

💬 多功能对话

📖 RAG 对话

🗃️ 知识库管理

工具设置

会话设置

☒ 启用Agent ⓘ

☐ 显示 Agent 过程

选择工具

股票分析报告生... ✕

✕ ▼

上传图片

Drag and drop file here

Limit 200MB per file • BMP, JPG, JPEG, PNG

Browse files

黏贴图像



帮我分析下这只股票：600519.SH，新闻页面是：

https://k.sina.com.cn/article_1747383115_6826f34b020020ar6.html



Financial Analysis Report for 600519.SH



1. Market Review

The stock 600519.SH has experienced mixed price performance over the past year. The 1-year price change is -6.27%, indicating a decline in value. However, in the last 3 months, the stock has seen a positive change of 6.18%, suggesting a recent recovery. Over the past month, the stock price has decreased by 4.27%. The latest closing price stands at 1565.94 yuan.

2. Risk Characteristics

The stock exhibits a 1-year annualized daily return volatility of 28.38%, which indicates a relatively high level of price fluctuation and risk. The historical maximum drawdown is -28.76%, reflecting the largest observed decline from a peak to a trough, highlighting significant downside risk.

3. Technical Indicators/Signals

Currently, there are no significant moving average crossover signals observed for 600519.SH. This



请输入对话内容，换行请使用Shift+Enter。



Financial Report System --- Q&A System

LangChain-Chatchat

当前版本: 0.3.1.3

多功能对话

RAG 对话

知识库管理

RAG 配置

会话设置

请选择对话模式:

知识库问答

请选择知识库:

Financial

历史对话轮数:

3

匹配知识条数:

3

知识匹配分数阈值:

0.03

0.002.00

Give me some information of AAPL Stock performance

知识库匹配结果

根据已知信息，AAPL（苹果公司）的股票价格为148.27美元，市值约为2.35万亿美元，市盈率为28.47。这些指标为投资者和利益相关者提供了关于公司市场价值的初步了解。

请输入对话内容，换行请使用Shift+Enter。

Financial Report System --- Q&A System

LangChain-Chatchat

当前版本: 0.3.1.3

多功能对话

RAG 对话

知识库管理

查询关键字

匹配条数

3

1100

添加文件到知识库

知识库 Financial 中已有文件:

知识库中包含源文件与向量库, 请从下表中选择文件后操作

序号	文档名称	文档加载器
3	LLM_judge_prompt.md	TextLoader
4	OPT_vs_RAW.md	TextLoader
5	600519SH-claude-3-5-sonnet-latest-fin-r1_evaluation.md	TextLoader
6	AAPL_fin-r1_FinReport.md	TextLoader
7	600519SH-llama-3.1-instruct-fin-r1_evaluation.md	TextLoader
8	AAPL_gpt-4o_FinReport.md	TextLoader
9	AAPL_claude-3-5-sonnet-latest_FinReport.md	TextLoader

Page Size: 1 to 10 of 27

Page 1 of 3

下载选中文档

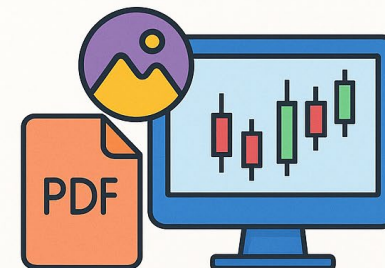
添加至向量库

从向量库删除

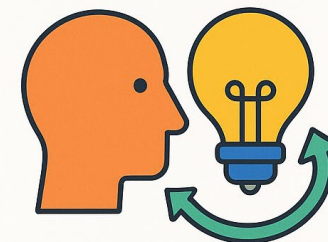
从知识库中删除

Conclusion and Future Work

- Financial Agent
 - Financial Report Agent
 - Refine Agent
 - LLM Judge
- Financial Report System
 - Chatbot
 - RAG-based Q&A
 - Knowledge Base Management



MULTIMODAL EXTENSIONS



**CONTINUAL LEARNING
MECHANISMS**



CROSS-ASSET ANALYSIS